Florian Frick

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Education

Master of Science in Computer Science *Aug 2023 — May 2025*
University of Colorado, Boulder — GPA 3.94

Bachelor of Science in Computer Science *Aug 2020 — May 2024*
University of Colorado, Boulder — GPA 3.87
Certificate of Engineering Leadership

Experience

Trimble Inc. Westminster, CO
Student Consultant for Capstone Project *Aug 2023 — May 2024*

- Led a team of 5 in creating an automatic calibration procedure for the 3D radar system of Trimble Inc.'s autonomous construction vehicles to reduce calibration time and eliminate the need for an on-site expert.
- Used Python and ROS2 to implement point clustering and superimposition to extract features and perform point alignment on noisy radar data to achieve a transformation error of down to 0.1m.
- Followed Agile methodology, using Jira, Docker, and Github for CI/CD and collaborative development.

Projects

NBA Fan Sentiment Dashboard *2026*

- Engineered a full-stack analytics platform using React, DynamoDB, and AWS AppSync to visualize trends and correlations between player statistics and fan sentiment.
- Built an automatic pipeline to scrape Reddit threads, perform one-shot sentiment analysis using Gemini's large language models, and make API requests for real-time updates of player performance.
- Created an agentic RAG AI chat feature using WebSockets and AWS Lambda that can answer complex user questions by combining real-time data retrieval with generative AI capabilities. Implemented conversation history and thought process streaming for improved user experience and transparency.

Imitation Learning for Robot Object Manipulation *2025*

- Used PyTorch to implement and train an encoder-decoder transformer model from scratch to maneuver a robot to push a T-shaped block with precision to a target location.
- Matched the performance of the state of the art Diffusion Policy model of 78% task completion while using 30% fewer parameters.

Autonomous Vehicle Competition Winner *2025*

- Used ROS2 and Python to autonomously race a robot around a track and adhere to traffic signs using computer vision (including OpenCV) and PID control. Leveraged pub/sub messaging and UDP communication to build a low-latency modular system that supports many sensors, processing nodes, and controllers.
- Completed three laps of the course fastest (avg 4.5 ft/s) out of 10 teams and without any human intervention.

Automatic Arctic Sea Ice Mapping *2024*

- Finetuned a Hugging Face transformer model to generate accurate sea ice concentration maps from satellite images of the Hudson Bay with Python using semantic segmentation.
- Improved performance over existing U-net models from 83% to 89% pixel accuracy and 0.44 to 0.50 IoU.

Skills

Languages: Python, TypeScript, Java, Javascript, C#, C++, SQL
Technologies: AWS, PyTorch, LangChain, NodeJS, React, Docker, Git, Linux, Windows, ROS2, CUDA
Certificates: AWS Cloud Practitioner (November 2025)